THE FLATNESS PROPERTY P IS STABLE UNDER BASE CHANGE BY THE FINITE TYPE PROPERTIES Q AND P+R

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the implication R implies Q is used as a black-box in-source citation (lemma-finite-type-finite-type-red); this card does not depend on that lemma's own proof

Editorial note. The label lemma-base-change-flat-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-flat-axioms (in-source definition carrier): defines property P (flatness of the relevant localizations) on WAdm^Noeth
- Lemma lemma-finite-type-red and lemma-finite-type (in-source definition carriers): define properties Q and R
- Formal Spaces, Lemma formal-spaces-lemma-completed-tensor-product: Noetherianity of the completed tensor under Q
- Lemma lemma-Noetherian-finite-type-red (in-source): Q implies topologically finite type
- Algebra, Lemma algebra-lemma-completion-flat: the completion of a finite type map is flat
- standard completed-tensor quotient identity: A (tensor)_B C/J^n = (A (completed tensor)_B C)/J^n(...)
- a limit lemma: a J-adically complete module whose quotients M/J^n M are flat over C/J^n is flat over C

Source reference. Original lemma 00MB: sources/stacks-chow/tags.txt, source lines 817–817. Prerequisite pointer invoked at 2992; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GB4: sources/stacks-chow/tags.txt, source lines 20359–20359. Prerequisite pointer invoked at 2974; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0GBF: sources/stacks-chow/tags.txt, source lines 20370–20370. Prerequisite pointer invoked at 2964; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original remark 0GBG: sources/stacks-chow/tags.txt, source lines 20371–20371. Prerequisite pointer invoked at 2961; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC6: sources/stacks-restricted/pinned-original-source.tex, source lines 2749–2749. Prerequisite pointer invoked at 2982; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 04U5: sources/stacks-restricted/pinned-original-source.tex, source lines 2439–2439. Prerequisite pointer invoked at 2956; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC5: sources/stacks-restricted/pinned-original-source.tex, source lines 2738–2738. Prerequisite pointer invoked at 2975; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC2: sources/stacks-restricted/pinned-original-source.tex, source lines 2631–2631. Prerequisite pointer invoked at 2954; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC9: sources/stacks-restricted/pinned-original-source.tex, source lines 2890–2890. Prerequisite pointer invoked at 2953; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0912: sources/stacks-chow/tags.txt, source lines 11430–11430. Prerequisite pointer invoked at 2999; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Denote P the property of arrows of $WAdm^{Noeth}$ defined in Lemma 0GC9. Denote Q the property defined in Lemma 0GC2 viewed as a property of arrows of $WAdm^{Noeth}$. Denote R the property defined in Lemma 04U5 viewed as a property of arrows of $WAdm^{Noeth}$. Then*

1. *P is stable under base change by Q (Formal Spaces, Remark 0GBG), and*
2. *$P + R$ is stable under base change (Formal Spaces, Remark 0GBF).*

Proof. The statement makes sense as each of the properties P, Q, and R is a local property of morphisms of $WAdm^{Noeth}$. Let $\varphi : B \to A$ and $\psi : B \to C$ be morphisms of $WAdm^{Noeth}$. If either $Q(\varphi)$ or $Q(\psi)$ then we see that $A \widehat{\otimes}_B C$ is Noetherian by Formal Spaces, Lemma 0GB4. Since R implies Q (Lemma 0GC5), we find that this holds in both cases (1) and (2). This is the first thing we have to check. It remains to show that $C \to A \widehat{\otimes}_B C$ is flat.

Proof of (1). Fix ideals of definition $I \subset A$ and $J \subset B$. By Lemma 0GC6 the ring map $B \to C$ is topologically of finite type. Hence $B \to C/J^n$ is of finite type for all $n \geq 1$. Hence $A \otimes_B C/J^n$ is Noetherian as a ring (because it is of finite type over A and A is Noetherian). Thus the I-adic completion $A \widehat{\otimes}_B C/J^n$ of $A \otimes_B C/J^n$ is flat over C/J^n because $C/J^n \to A \otimes_B C/J^n$ is flat as a base change of $B \to A$ and because $A \otimes_B C/J^n \to A \widehat{\otimes}_B C/J^n$ is flat by Algebra, Lemma 00MB Observe that $A \widehat{\otimes}_B C/J^n = (A \widehat{\otimes}_B C)/J^n (A \widehat{\otimes}_B C)$; details omitted. We conclude that $M = A \widehat{\otimes}_B C$ is a C-module which is complete with respect to the J-adic

topology such that M/J^nM is flat over C/J^n for all $n \geq 1$. This implies that M is flat over C by More on Algebra, Lemma 0912.

Proof of (2). In this case $B \to A$ is adic and hence we have just $A \widehat{\otimes}_B C = \lim A \otimes_B C/J^n$. The rings $A \otimes_B C/J^n$ are Noetherian by an application of Formal Spaces, Lemma 0GB4 with C replaced by C/J^n. We conclude in the same manner as before. □